Toulik Maitra

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Education

University of California, Davis | Ph.D., Chemical Engineering — Sep 2022 – Jun 2027

- GPA 4.0/4.0. *Awards:* UC Davis Dissertation Fellowship (top 1%, 2026); Graduate Research Award ($3,000, 2025); Max Planck Fellowship ($3,500, 2025); Outstanding Poster, CHMS Symposium (2023).

University of Calcutta | B.Tech., Chemical Engineering — Aug 2016 – Aug 2020

- CGPA 8.36/10.0. *Awards:* Best Paper (IIT Kharagpur); Most Innovative Idea in Nanoscience (IIT Bombay); Top 10, Global Entrepreneurship Challenge (IIT Kharagpur).

Professional Experience

Visiting Fellow | *Max Planck Institute for Polymer Research, Germany* — Jul 2025 – Present

- Partnered with four groups in Paul Blom's Department of Molecular Electronics to deliver pre-fabrication material selection tools; tools adopted into VOTCA's public release.
- Developed models that let the Blom group prioritize OLED candidates before fabrication; improved prediction accuracy by 60%, cut screening time by 30%.
- Co-authored three papers, one under review (2026).

Graduate Researcher | *University of California, Davis* — Jan 2023 – Present

- Served as computational liaison between Oak Ridge National Laboratory and Iowa State University; translated model outputs into design recommendations that shaped experimental priorities for partner teams.
- Developed models that connect molecular arrangement to electrical conductivity; improved prediction accuracy by 40% over prior benchmarks.
- First-authored paper under review at *Advanced Functional Materials* (2026); contributed to three additional publications, including *Chemistry of Materials* (2024).

Business Development Intern | *TATA Autocomp Systems Ltd, Pune, India* — May 2019 – Jul 2019

- Partnered with Marketing, Sales, and Engineering stakeholders to evaluate alternative engine hood designs; built a cost-performance model across material and manufacturing dimensions.
- Identified a $100K cost reduction; presented executive recommendation to VP-level commercial and engineering leadership; ranked top 10% of intern cohort.

Undergraduate Researcher | *University of Calcutta, Kolkata, India* — Aug 2016 – Dec 2021

- Ran experiments on light-absorbing nanomaterials with researchers at Jadavpur University and IIT Kharagpur; identified a low-cost nanomaterial candidate as a viable alternative to platinum catalysts.
- Published 6 peer-reviewed papers on how nanomaterial structure controls light absorption and electrical performance.

Leadership Experience

- **Research Mentor, E-Search (UC Davis, 2023–present):** Supervised three undergraduates on molecular simulation models; each co-authored a publication and secured a NASA internship.
- **Teaching Assistant (UC Davis, Apr 2023–Mar 2025):** Taught nine courses (1,000+ students total) in thermodynamics, transport, and materials processing; received average student rating of 4.7/5.
- **Co-Founder, eCell (Univ. of Calcutta, 2018–2020):** Built the entrepreneurship cell from scratch; mentored students who secured PwC associate roles and MBA admissions.

Selected Publications (2 of 10)

- **T. Maitra** *et al.* Phonon Spectra Reveal How Selective Fluorination Enhances Hole Transport in Benzodithiophenes. *Under review* (2026).
- F. Post, . . ., **T. Maitra** *et al.* Ab initio parametrization of distributed polarizable force fields. arXiv:2606.09647 (2026).

Skills & Interests

Technical: PowerPoint, Excel, Python, MATLAB; machine learning (regression, neural networks), Bayesian inference, Monte Carlo simulation; quantitative modeling and finite element analysis.

Languages: English (fluent), Bengali (native), Hindi (conversational).

Interests: Soccer attacking midfielder (UC Davis intramural champion; Manchester United fan) and cricket all-rounder. Run 10K. Cook Mexican and regional Indian. Midway through *Educated* by Tara Westover. F1 fan (Verstappen). Booked Prague on 24 hours' notice.